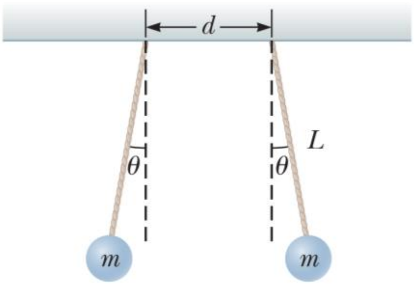
## 1. class, Electrodynamics and Optics

**Page 610, Task 12.**

Particle A of charge 3.00$\times$ $10^{-4}$ C is at the origin, particle B of charge -6.00$\times$ $10^{-4}$ C is at (4.00 m, 0), and particle C of charge 1.00$\times$ $10^{-4}$ C is at (0, 3.00 m). We wish to find the net electric force on C.
(a) What is the x component of the electric force exerted by A on C?
(b) What is the y component of the force exerted by A on C?
(c) Find the magnitude of the force exerted by B on C.
(d) Calculate the x component of the force exerted by B on C.
(e) Calculate the y component of the force exerted by B on C.
(f) Sum the two x components from parts (a) and (d) to obtain the resultant x component of the electric force acting on C.
(g) Similarly, find the y component of the resultant force vector acting on C.
(h) Find the magnitude and direction of the resultant electric force acting on C.

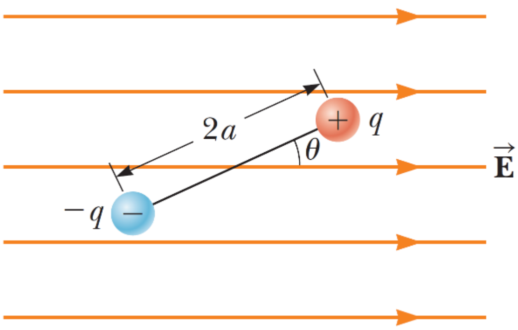
**Page 613, Task 43.**

Two hard rubber spheres, each of mass $m = 15.0$ g, are rubbed with fur on a dry day and are then suspended with two insulating strings of length $L = 5.00$ cm whose support points are a distance $d = 3.00$ cm from each other as shown in Figure P22.43. During the rubbing process, one sphere receives exactly twice the charge of the other. They are observed to hang at equilibrium, each at an angle of $\theta = 10.0°$ with the vertical. Find the amount of charge on each sphere.




**Page 614, Task 50.**

An electric dipole in a uniform horizontal electric field is displaced slightly from its equilibrium position as shown in Figure P22.50, where $\theta$ is small. The separation of the charges is $2a$, and each of the two particles has mass $m$.




(a) Assuming the dipole is released from this position, show that its angular orientation exhibits simple harmonic motion with a frequency

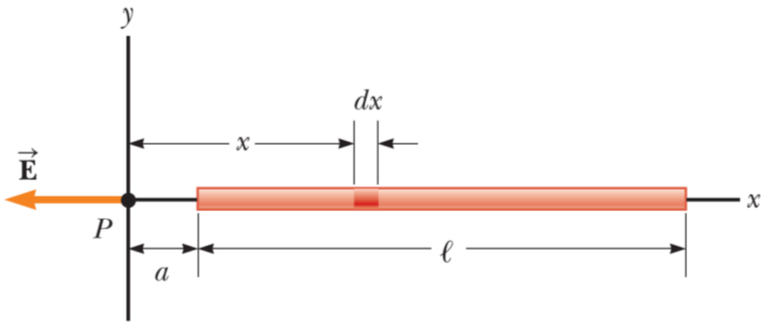
$$f = \frac{1}{2\pi}\sqrt{\frac{qE}{ma}}.$$

**(optional)** What If? (b) Suppose the masses of the two charged particles in the dipole are not the same even though each particle continues to have charge $q$. Let the masses of the particles be $m_1$ and $m_2$. Show that the frequency of the oscillation in this case is

$$f = \frac{1}{2\pi}\sqrt{\frac{qE(m_1 + m_2)}{2am_1m_2}}.$$

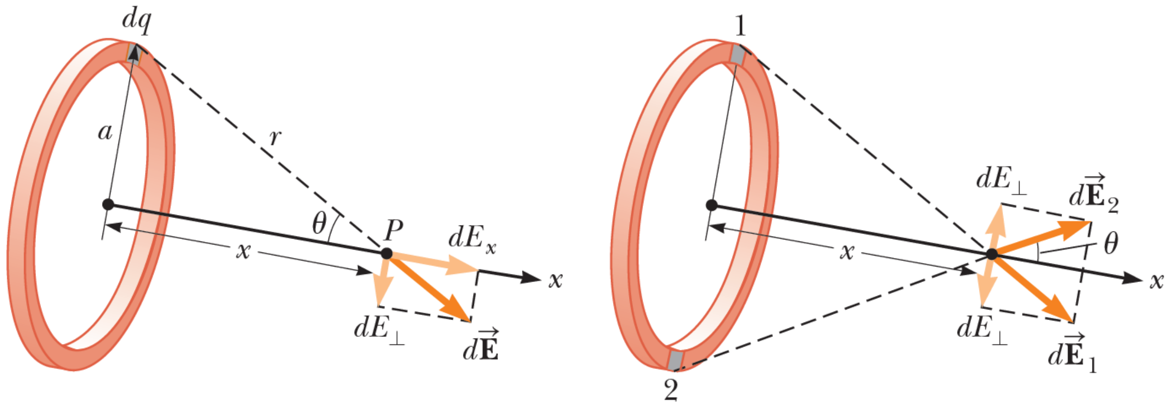
**Page 618, Task 23.1.**

A rod of length $l$ has a uniform positive charge per unit length $\lambda$ and a total charge $Q$. Calculate the electric field at a point $P$ that is located along the long axis of the rod and a distance $a$ from one end (Fig. 23.2).




**Page 618, Task 23.2.**

A ring of radius $a$ carries a uniformly distributed positive total charge $Q$. Calculate the electric field due to the ring at a point $P$ lying a distance $x$ from its center along the central axis perpendicular to the plane of the ring (Fig. 23.3a).




**Page 620, Task 23.3.**

A disk of radius $R$ has a uniform surface charge density $\sigma$. Calculate the electric field at a point $P$ that lies along the central perpendicular axis of the disk and a distance $x$ from the center of the disk (Fig. 23.4).